Rešitve 1. kol 2 naloga

Monday, 21 December 2020 14:40

2.) $u_L = f(x-ct) + g(x+ct)$ (2t) [a) (8t)]

$u_D = \varphi(x-ct)$

Zveznost pri $x=0$ $\quad f(-ct) + g(ct) = \varphi(-ct)$

$\hookrightarrow g(z) = \varphi(-z) - f(-z)$ (1)

2NZ za maso

$$F\left(-\frac{\partial u_L}{\partial x} + \frac{\partial u_D}{\partial x}\right)\Big|_{x=0} = m\ddot{u}_D\Big|_{x=0}$$

$$F(-f'(-ct) - g'(ct) + \varphi'(-ct)) = mc^2\varphi''(-ct)$$ (2) (4t)

iz (1) in (2) dobimo

$$-\frac{mc^2}{2F}\varphi'' + \varphi' = f' \quad \int$$ (2t)

$$\boxed{-\frac{mc^2}{2F}\varphi' + \varphi = f}$$ (konstante ne dobimo)

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b) $-\varphi' + \varphi = z^{2a}e^z$ [b) (3t)]

nastavek $\varphi = A z^\alpha e^z \to (-\alpha z^{\alpha-1} - z^\alpha + z^\alpha)Ae^z = z^{2a}e^z$

$\to \alpha = 2a+1$, $A = -\frac{1}{\alpha}$

$$\boxed{\varphi = -\frac{z^{2a+1}}{(2a+1)}e^z}, \quad z<0$$ (vse skupaj je pozitivno)

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c) $\varphi' = 0 \Rightarrow z = -(2a+1) = x - ct$

$$\underbrace{t = \frac{2a+1}{c}}$$

c) (1 t)